

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 08

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 For which values of x is $\sqrt{x^2} = -x$?

A $x = 0$ only

B $x \geq 0$

C $x \leq 0$

D $x < 0$ only

E all real x

F no real x

2 The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .

A $k = \pm 2\sqrt{2}$

B $k = \pm 8$

C $k = \pm 4$

D $k = \pm \sqrt{8}$

E $k = 4$ only

F $k = \pm 2$

3 The quadratic $y = ax^2 + bx + c$ has its vertex at $(2, -3)$ and passes through $(0, 5)$. Find $a + b + c$.

A -1

B 7

C 1

D 5

E -3

F 3

- 4 Which of the following is the negation of '**there exists a positive integer n such that $n^2 + n + 41$ is not prime**'?
- A For every positive integer n , $n^2 + n + 41$ is not prime.
 - B There exists a positive integer n such that $n^2 + n + 41$ is prime.
 - C For every positive integer n , $n^2 + n + 41$ is prime.
 - D For some positive integer n , $n^2 + n + 41$ is prime.
 - E There is no positive integer n for which $n^2 + n + 41$ is prime.

5 Claim: for all real x , if $x^2 > 9$ then $x > 3$.

I Suppose $x^2 > 9$.

II Taking square roots of both sides, $x > 3$.

III This proves the claim.

Which of the following is true?

A The claim is false; the first error is on line II, since $x = -4$ satisfies $x^2 > 9$ but not $x > 3$.

B The claim and proof are both correct.

C The claim is true but line II should say $|x| > 3$ and then deduce $x > 3$.

D The claim is false; the first error is on line I.

E The claim is true; the first error is on line III.

6 Here is a proof that $\sqrt{2}$ is irrational, with its steps in the wrong order.

I Then $p^2 = 2q^2$, so p^2 is even, so p is even; write $p = 2k$.

II Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ where p and q are positive integers with no common factor.

III This contradicts the assumption that p and q have no common factor, so $\sqrt{2}$ is irrational.

IV Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.

Which is the correct order?

A II, I, IV, III

B I, II, IV, III

C II, IV, I, III

D I, IV, II, III

E II, I, III, IV

7 Which of the following statements about the graph of $y = |2x - 4|$ is/are true?

- I** It meets the x -axis at exactly one point.
- II** It is symmetric about the line $x = 2$.
- III** It lies entirely on or above the line $y = 0$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 8** The equation $x^2 + bx + c = 0$ has real coefficients. Which of the following is a **sufficient** condition for it to have two distinct real roots?

- A** $b \neq 0$
- B** $b > 0$
- C** $c > 0$
- D** $b^2 \geq 4c$
- E** $c < 0$

- 9** Let P be the statement ' $x > 2$ ' and Q the statement ' $x^2 > 4$ ', where x is a real number. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

- 10** A proof that **if a and b are positive real numbers then $\frac{a+b}{2} \geq \sqrt{ab}$ ** is given with its lines scrambled.

I Hence $a + b \geq 2\sqrt{ab}$, and dividing by 2 gives the result.

II Expanding, $a - 2\sqrt{ab} + b \geq 0$.

III Since squares are non-negative, $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

Which order gives a valid proof?

A III, II, I

B I, II, III

C II, III, I

D III, I, II

E II, I, III

11 Consider the statement: ****if a and b are irrational, then $a + b$ is irrational.**** Which of the following pairs is a counterexample?

A $a = 1, b = -1$

B $a = \sqrt{2}, b = -\sqrt{2}$

C $a = \pi, b = \sqrt{2}$

D $a = \sqrt{2}, b = 2\sqrt{2}$

E $a = \sqrt{2}, b = \sqrt{3}$

12 Here is a purported proof that $1 = 2$.

I Let $a = b$, with $a \neq 0$.

II Then $a^2 = ab$.

III So $a^2 - b^2 = ab - b^2$, i.e. $(a - b)(a + b) = b(a - b)$.

IV Dividing both sides by $(a - b)$: $a + b = b$.

V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

A IV

B II

C III

D V

E I

- 13** It is given that for all real x : if $f(x) > 0$ then $g(x) > 0$. It is also known that $g(2) = -1$. What can be deduced?
- A** $f(2) < 0$
 - B** $f(2) = 0$
 - C** $f(2) > 0$
 - D** $f(2) \leq 0$
 - E** nothing about $f(2)$

- 14** Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I** The circles touch externally.
- II** The circles intersect at exactly two points.
- III** The point $(5, 0)$ lies on both circles.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

15 n is a positive integer such that n , $n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I** There is exactly one such value of n .
- II** n must be odd.
- III** One of n , $n + 2$, $n + 4$ must be divisible by 3.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 16** Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?
- A** Ben
 - B** Alice
 - C** Cara
 - D** It cannot be determined.
 - E** The situation is impossible.

17 Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

18 Given that $|2x - 3| < 5$, which of the following statements is/are true?

I $-1 < x < 4$

II $x^2 < 16$

III $x > -1$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

19 A selection of n distinct integers is taken from $1, 2, \dots, 20$. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

A 2

B 15

C 12

D 11

E 20

F 10

20 $f(x) = x^2 + 1$. Which of the following statements is/are true?

I $f(f(x)) = x^4 + 2x^2 + 2$ for all x

II $f(x + 1) = f(x) + 1$ for all x

III $f(2x) = 4f(x) - 3$ for all x

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

END OF TEST